

1. (a) $\vec{E}_1 = \hat{n} E_0 \cos(kz - \omega t)$; $\vec{E}_2 = \hat{n} E_0 \cos(kz + \omega t)$
 $\therefore \vec{E} = \vec{E}_1 + \vec{E}_2 = \hat{n} E_0 [\cos(kz - \omega t) + \cos(kz + \omega t)]$
 $= \hat{n} [\cancel{\cos kz \cos \omega t} + \cancel{\sin kz \sin \omega t} + \cancel{\cos kz \cos \omega t} - \cancel{\sin kz \sin \omega t}]$
 $= 2\hat{n} \cos kz \cos \omega t.$

This is a standing wave.

(b) Direction of propagation for $\vec{E}_1$ is $+\hat{z}$
 for $\vec{E}_2$ is $-\hat{z}$

$\vec{E}_1$ points in $\hat{n}$ direction. & $\vec{E}_2$ points in the $\hat{n}$ direction.

The travelling magnetic fields must point in a direction $\perp^r$ to both the associated $\vec{E}$ field and the direction of propagation.

$\therefore \vec{B}_1$ points in $+\hat{y}$ direction

$\vec{B}_2$ points in $-\hat{y}$ direction.

Further, $B_{10} = \frac{E_0}{c}$ & $B_{20} = \frac{E_0}{c}.$

$$\therefore \vec{B}_1 = \hat{y} \left(\frac{E_0}{c} \right) \cos(kz - \omega t)$$

$$\vec{B}_2 = -\hat{y} \left(\frac{E_0}{c} \right) \cos(kz + \omega t)$$

$$\vec{B} = \vec{B}_1 + \vec{B}_2 = \hat{y} \frac{E_0}{c} [\cos(kz - \omega t) - \cos(kz + \omega t)]$$

$$= \hat{y} \frac{2E_0}{c} \sin kz \sin \omega t.$$

(c) Maxwell's eq: $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

$$\vec{\nabla} \times \vec{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2E_0 \cos kx \cos \omega t & 0 & 0 \end{vmatrix}$$

$$= +\hat{y} \frac{\partial}{\partial z} (2E_0 \cos kx \cos \omega t)$$

$$= -\hat{y} 2E_0 k \sin kx \cos \omega t$$

$$\therefore \frac{\partial \vec{B}}{\partial t} = \hat{y} 2E_0 k \sin kx \cos \omega t$$

Integrating, $\vec{B} = \hat{y} 2E_0 \left(\frac{k}{\omega}\right) \sin kx \sin \omega t$

$$= \hat{y} 2 \frac{E_0}{c} \sin kx \sin \omega t$$

since, $\omega = ck$

Note: we have ignored the constant of integration.

2. $y(x, t) = y_0 \cos(kx - \omega t)$

$$v = 20/\text{s}; v = 80 \text{ m/s}$$

$$\omega = 2\pi v = 40\pi \text{ rad/s} \quad \& \quad k = \frac{\omega}{v} = \frac{40\pi}{80} = \frac{\pi}{2} \text{ rad/m}$$

(e) Now, phase difference: $\delta = (kx_2 - \omega t) - (kx_1 - \omega t)$
 $= k(x_2 - x_1)$

Given that displacements are 30° apart in phase $\Rightarrow$ i.e. $\frac{\pi}{6}$ rad apart $\Rightarrow$ phase.

$$\therefore k(x_2 - x_1) = \frac{\pi}{6} \Rightarrow \frac{\pi}{2} (x_2 - x_1) = \frac{\pi}{6}$$

$$\Rightarrow x_2 - x_1 = \frac{1}{3} \text{ m}$$

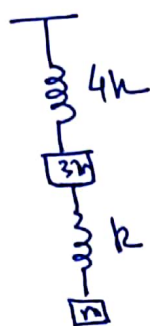
(b) When they are separated in times

$$\delta = (kx - \omega t_2) - (kx - \omega t_1)$$

$$= k\omega(t_1 - t_2) = 40\pi \times 0.01$$

$$= 0.4\pi = \frac{2\pi}{5} \approx 72^\circ$$

3.



$$3m \frac{d^2 x_1}{dt^2} = -4kx_1 + k(x_2 - x_1) = -5kx_1 + kx_2$$

$$m \frac{d^2 x_2}{dt^2} = -k(x_2 - x_1)$$

$$x_1 = A e^{i\omega t}, \quad x_2 = B e^{i\omega t}$$

Then, $-3m\omega^2 A = -5kA + kB$

$$\Rightarrow -\omega^2 A = -\frac{5k}{3m} A + \frac{k}{3m} B \quad \text{--- (1)}$$

$$-m\omega^2 B = -k(B - A)$$

$$\Rightarrow -\omega^2 B = -\frac{k}{m} B + \frac{k}{m} A \quad \text{--- (2)}$$

$$\therefore -\omega^2 \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} -5k/3m & k/3m \\ k/m & -k/m \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix}$$

$$\Rightarrow \begin{bmatrix} -\frac{5k}{3m} + \omega^2 & k/3m \\ \frac{k}{m} & -\frac{k}{m} + \omega^2 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = 0$$

For non-trivial solution,

$$\begin{vmatrix} -\frac{5k}{3m} + \omega^2 & \frac{k}{3m} \\ \frac{k}{m} & -\frac{k}{m} + \omega^2 \end{vmatrix} = 0$$

$$\Rightarrow \left(-\frac{5k}{3m} + \omega^2\right)\left(-\frac{k}{m} + \omega^2\right) - \frac{k^2}{3m} = 0$$

$$\Rightarrow \omega^4 - \left(\frac{5k}{3m} + \frac{k}{m} \right) \omega^2 + \frac{5k^2}{3m^2} - \frac{k^2}{3m^2} = 0$$

$$\Rightarrow \omega^4 - \frac{8k}{3m} \omega^2 + \frac{4k^2}{3m^2} = 0$$

$$\text{Let, } \omega^2 = X \quad \therefore X^2 - \frac{8k}{3m} X + \frac{4k^2}{3m^2} = 0$$

$$X = \frac{\frac{8k}{3m} \pm \sqrt{\frac{64k^2}{9m^2} - \frac{16k^2}{3m^2}}}{2}$$

$$= \frac{\frac{8k}{3m} \pm \sqrt{\frac{16k^2}{9m^2}}}{2}$$

$$= \frac{\frac{8k}{3m} \pm \frac{4k}{3m}}{2}$$

$$\therefore \omega^2 = \frac{2k}{m}, \frac{2k}{3m}$$

$$\Rightarrow \omega = \sqrt{\frac{2k}{m}}, \sqrt{\frac{2k}{3m}}$$

(b) Substituting for $\omega^2 = \frac{2k}{m}$ in (1),

$$-\frac{2k}{m} A = -\frac{5k}{3m} A + \frac{k}{3m} B$$

$$\Rightarrow -\frac{k}{3m} A = \frac{k}{3m} B \Rightarrow A = -B$$

$$\Rightarrow B = -A$$

At any instance masses are equidistant from their equilibrium positions & are on opposite sides of them.

Substituting for $\omega^2 = \frac{2k}{3m}$ in (1),

$$-\frac{2k}{3m}A = -\frac{5k}{3m}A + \frac{k}{3m}B.$$

$$\Rightarrow \frac{k}{m}A = \frac{k}{3m}B \Rightarrow B = 3A.$$

At any instance, the masses are both either above or below their equilibrium positions, the displacement of the lower mass being 3 times that of the higher mass.

4. $\mu_1 = 1.0 \text{ g/cm} ; \mu_2 = 4.0 \text{ g/cm}.$

$A_1 = 3 \text{ cm} ; \lambda_1 = 25 \text{ cm}.$

(a) $v = \sqrt{\frac{T}{\mu}}$ & freq. $v = \frac{v}{\lambda}$.

Frequency is the same in both string.

$$\therefore \frac{v_1}{\lambda_1} = \frac{v_2}{\lambda_2} \Rightarrow \sqrt{\frac{T}{\mu_1}} \cdot \frac{1}{\lambda_1} = \sqrt{\frac{T}{\mu_2}} \cdot \frac{1}{\lambda_2}$$

$$\Rightarrow \frac{\lambda_2}{\lambda_1} = \sqrt{\frac{\mu_1}{\mu_2}} = \sqrt{\frac{1}{4}} = \frac{1}{2}.$$

$$\Rightarrow \lambda_2 : \lambda_1 = \frac{25}{2} = 12.5 \text{ cm}.$$

Further, $\frac{A_2}{A_1} = \frac{2v_2}{v_1 + v_2} = \frac{2\sqrt{T/\mu_2}}{\sqrt{T/\mu_1} + \sqrt{T/\mu_2}}$

$$= \frac{2\sqrt{\mu_1}}{\sqrt{\mu_1} + \sqrt{\mu_2}} = \frac{2}{1 + \sqrt{\frac{\mu_2}{\mu_1}}}$$

$$= \frac{2}{1 + \sqrt{4}} = \frac{2}{1 + 2} = \frac{2}{3}.$$

$$\therefore A_2 = \frac{2}{3}A_1 = \frac{2}{3} \cdot 3 = 2 \text{ cm}.$$

$$\begin{aligned}
 (b) \quad \frac{A_r}{A_i} &= \frac{v_2 - v_1}{v_2 + v_1} = \frac{\sqrt{T/\mu_2} - \sqrt{T/\mu_1}}{\sqrt{T/\mu_2} + \sqrt{T/\mu_1}} \\
 &= \frac{\sqrt{\mu_1} - \sqrt{\mu_2}}{\sqrt{\mu_1} + \sqrt{\mu_2}} = \frac{1 - \sqrt{\mu_2/\mu_1}}{1 + \sqrt{\mu_2/\mu_1}} \\
 &= \frac{1 - 2}{1 + 2} = -\frac{1}{3}
 \end{aligned}$$

Fraction of power reflected would be the ratio of the flux for the reflected & incident waves.

$$(\bar{E}_f)_r = \frac{1}{2} \mu_1 k_{oi} \tilde{v}_1 = \frac{1}{2} \mu_1 \omega^2 A_r^2 v_1$$

$$(\bar{E}_f)_i = \frac{1}{2} \mu_1 k_{oi} \tilde{v}_1 = \frac{1}{2} \mu_1 \omega^2 A_i^2 v_1$$

$\therefore$ Fraction of power reflected is

$$= \frac{(\bar{E}_f)_r}{(\bar{E}_f)_i} = \left(\frac{A_r}{A_i} \right)^2 = \frac{1}{9}$$

5. In complex notation,

$$\vec{E}_i = \tilde{E}_{oi} e^{i(\vec{k}_i \cdot \vec{r} - \omega t)} \hat{y}$$

$$\vec{B}_i = \frac{1}{v_1} \tilde{E}_{oi} e^{i(\vec{k}_i \cdot \vec{r} - \omega t)} (-\cos\theta_i \hat{x} + \sin\theta_i \hat{z})$$

$$\vec{E}_r = \tilde{E}_{or} e^{i(\vec{k}_r \cdot \vec{r} - \omega t)} \hat{y}$$

$$\vec{B}_r = \frac{\tilde{E}_{or}}{v_1} e^{i(\vec{k}_r \cdot \vec{r} - \omega t)} (\cos\theta_i \hat{x} + \sin\theta_i \hat{z})$$

$$\vec{E}_t = \tilde{E}_{ot} e^{i(\vec{k}_t \cdot \vec{r} - \omega t)} \hat{y}$$

$$\vec{B}_t = \frac{\tilde{E}_{ot}}{v_2} e^{i(\vec{k}_t \cdot \vec{r} - \omega t)} (-\cos\theta_t \hat{x} + \sin\theta_t \hat{z})$$

Boundary conditions:

$$(i) \epsilon_1 E_1^\perp = \epsilon_2 E_2^\perp \quad (ii) \vec{E}^\parallel = \vec{E}^\parallel$$

$$(iii) B_1^\perp = B_2^\perp \quad (iv) \frac{1}{\mu_1} \vec{B}_1^\parallel = \frac{1}{\mu_2} \vec{B}_2^\parallel$$

Law of refraction gives, $\frac{\sin\theta_t}{\sin\theta_i} = \frac{n_1}{n_2} = \frac{v_2}{v_1}$

$$\& \vec{k}_i \cdot \vec{r} - \omega t = \vec{k}_r \cdot \vec{r} - \omega t = \vec{k}_t \cdot \vec{r} - \omega t \text{ at } z=0$$

$\therefore$ All exponential factors can be dropped when applying the boundary conditions.

(i) gives, $0 = 0$.

$$(ii) \text{ gives, } \frac{1}{v_1} \tilde{E}_{oi} \sin\theta_i + \frac{1}{v_1} \tilde{E}_{or} \sin\theta_i = \frac{1}{v_2} \tilde{E}_{ot} \sin\theta_t$$

$$\Rightarrow \tilde{E}_{oi} + \tilde{E}_{or} = \frac{v_1 \sin\theta_t}{v_2 \sin\theta_i} \tilde{E}_{ot} = \tilde{E}_{ot} \quad \text{---(a)}$$

$$(iii) \text{ gives, } \tilde{E}_{oi} + \tilde{E}_{or} = \tilde{E}_{ot} \text{ (same as above)} \quad \text{---(b)}$$

$$(iv) \text{ Also, } \frac{1}{\mu_1} \left[\frac{1}{v_1} \tilde{E}_{oi} (-\cos \theta_i) + \frac{1}{v_1} \tilde{E}_{or} \cos \theta_i \right] = \frac{1}{\mu_2 v_2} \tilde{E}_{ot} (-\cos \theta_t)$$

$$\Rightarrow \tilde{E}_{oi} - \tilde{E}_{or} = \left(\frac{\mu_1 v_1 \cos \theta_i}{\mu_2 v_2 \cos \theta_i} \right) \tilde{E}_{or}$$

$$\text{Let } \alpha = \frac{\cos \theta_t}{\cos \theta_i} \quad \& \quad \beta = \frac{\mu_1 v_1}{\mu_2 v_2}$$

$$\therefore, \tilde{E}_{oi} - \tilde{E}_{or} = \alpha \beta \tilde{E}_{or} \quad \text{--- (c)}$$

$$\therefore \text{ From (a) \& (c), } 2\tilde{E}_{oi} = (1 + \alpha \beta) \tilde{E}_{or}$$

$$\Rightarrow \tilde{E}_{or} = \frac{2}{1 + \alpha \beta} \tilde{E}_{oi}$$

$$\& \tilde{E}_{ot} = \tilde{E}_{or} - \tilde{E}_{oi} = \left(\frac{2}{1 + \alpha \beta} - \frac{1 + \alpha \beta}{1 + \alpha \beta} \right) \tilde{E}_{oi}$$

$$= \left(\frac{1 - \alpha \beta}{1 + \alpha \beta} \right) \tilde{E}_{oi}$$

$\therefore$ In terms of real amplitudes:

$$\boxed{E_{ot} = \frac{2}{1 + \alpha \beta} E_{oi} ; E_{or} = \left| \frac{1 - \alpha \beta}{1 + \alpha \beta} \right| E_{oi}}$$

$\Rightarrow$ Find equations for polarisation $\perp$ to the plane of incidence.

$$(b) R = \frac{I_r}{I_i} = \left(\frac{E_{or}}{E_{oi}} \right)^2 = \left(\frac{1 - \alpha \beta}{1 + \alpha \beta} \right)^2$$

$$T = \frac{I_t}{I_i} = \frac{\epsilon_2 v_2}{\epsilon_1 v_1} \left(\frac{E_{ot}}{E_{oi}} \right)^2 \frac{\cos \theta_t}{\cos \theta_i} = \alpha \beta \left(\frac{2}{1 + \alpha \beta} \right)^2$$

$$\therefore R + T = \frac{(1 - \alpha \beta)^2 + 4\alpha \beta}{(1 + \alpha \beta)^2} = \frac{(1 + \alpha \beta)^2}{(1 + \alpha \beta)^2} = 1$$

$$6. \quad y(x,t) = \sum_n \left[A_n \cos(\omega_n t) + B_n \sin(\omega_n t) \right] \sin\left(\frac{n\pi}{L}x\right).$$

$$A_n = \frac{2}{L} \int_0^L y(x,0) \sin\left(\frac{n\pi}{L}x\right) dx$$

$$B_n = \frac{2}{\omega_n L} \int_0^L \dot{y}(x,0) \sin\left(\frac{n\pi}{L}x\right) dx$$

$$\text{At } t=0, \quad y(x,0) = \begin{cases} h \frac{x}{d} & \text{for } 0 \leq x \leq d \\ h \left(\frac{L-x}{L-d}\right) & \text{for } d \leq x \leq L. \end{cases}$$

$$\& \quad \dot{y}(x,0) = 0$$

$$\therefore B_n = 0.$$

$$A_n = \frac{2}{L} \left[\int_0^d h \frac{x}{d} \sin\left(\frac{n\pi}{L}x\right) dx + \int_d^L h \left(\frac{L-x}{L-d}\right) \sin\left(\frac{n\pi}{L}x\right) dx \right]$$

$$= \frac{2}{L} \left[\frac{h}{d} \int_0^d x \sin\left(\frac{n\pi}{L}x\right) dx + \frac{h}{L-d} \int_d^L (L-x) \sin\left(\frac{n\pi}{L}x\right) dx \right]$$

$$\int_0^d x \sin\left(\frac{n\pi}{L}x\right) dx = -x \frac{\cos\left(\frac{n\pi}{L}x\right)}{\frac{n\pi}{L}} \Big|_0^d + \int_0^d 1 \cdot \frac{\cos\left(\frac{n\pi}{L}x\right)}{\frac{n\pi}{L}} dx$$

$$= -\frac{L}{n\pi} \left[d \cos\left(\frac{n\pi d}{L}\right) - 0 \right] + \frac{L}{n\pi} \int_0^d \cos\left(\frac{n\pi}{L}x\right) dx$$

$$= -\frac{Ld}{n\pi} \cos\left(\frac{n\pi d}{L}\right) + \frac{L^2}{n^2\pi} \sin\left(\frac{n\pi d}{L}\right).$$

$$\int_d^L (L-x) \sin\left(\frac{n\pi x}{L}\right) dx = -\frac{(L-x)}{\frac{n\pi}{L}} \cos\left(\frac{n\pi x}{L}\right) \Big|_d^L$$

$$+ \int_d^L (-1) \cdot \frac{\cos\left(\frac{n\pi x}{L}\right)}{\frac{n\pi}{L}} dx$$

$$= -\frac{L}{n\pi} \left[0 - (L-d) \cos\left(\frac{n\pi d}{L}\right) \right]$$

$$= -\frac{L}{n\pi} \int_d^L \cos\left(\frac{n\pi x}{L}\right) dx$$

$$= + \frac{L(L-d)}{n\pi} \cos\left(\frac{n\pi d}{L}\right) - \frac{L^2}{n^2\pi^2} (0 - \sin\left(\frac{n\pi d}{L}\right))$$

$$= + \frac{L(L-d)}{n\pi} \cos\left(\frac{n\pi d}{L}\right) + \frac{L^2}{n^2\pi^2} \sin\left(\frac{n\pi d}{L}\right)$$

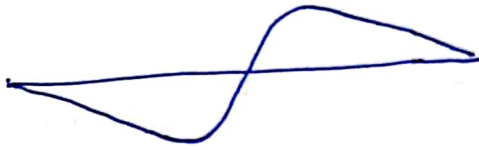
$$\therefore A_n = \frac{2}{L} \left[-\frac{Lh}{n\pi} \cos\left(\frac{n\pi d}{L}\right) + \frac{L^2h}{n^2\pi^2 d} \sin\left(\frac{n\pi d}{L}\right) \right. \\ \left. + \frac{Lh}{n\pi} \cos\left(\frac{n\pi d}{L}\right) + \frac{L^2h}{n^2\pi^2 (L-d)} \sin\left(\frac{n\pi d}{L}\right) \right]$$

$$= \frac{2}{L} \cdot \frac{2L^2h}{n^2\pi^2} \left(\frac{1}{d} + \frac{1}{L-d} \right) \sin\left(\frac{n\pi d}{L}\right)$$

$$= \frac{2L^2h}{n^2\pi^2 d(L-d)} \sin\left(\frac{n\pi d}{L}\right) = \frac{2h}{n^2\pi^2 d(L-d)} \frac{L^2}{L} \sin\left(\frac{n\pi d}{L}\right)$$

$$\therefore y(x,t) = \frac{2hL^2}{d(L-d)} \sum_n \frac{1}{n^2\pi^2} \sin\left(\frac{n\pi d}{L}\right) \sin\left(\frac{n\pi x}{L}\right) \cos(\omega_n t)$$

(b)



(c)

